

Complex wavelet filter improves FLIM phasors for photon starved imaging experiments: supplement

**P. WANG,¹ F. HECHT,² G. OSSATO,² S. TILLE,² S. E. FRASER,¹
AND J. A. JUNGE^{1,*} **

¹*Translational Imaging Center, Dornsife School of Letters, Arts, and Sciences, University of Southern California, 1002 Childs Way, Los Angeles, CA 90089, USA*

²*Leica Microsystems CMS GmbH, Am Friedensplatz 3, Mannheim 68165, Germany*

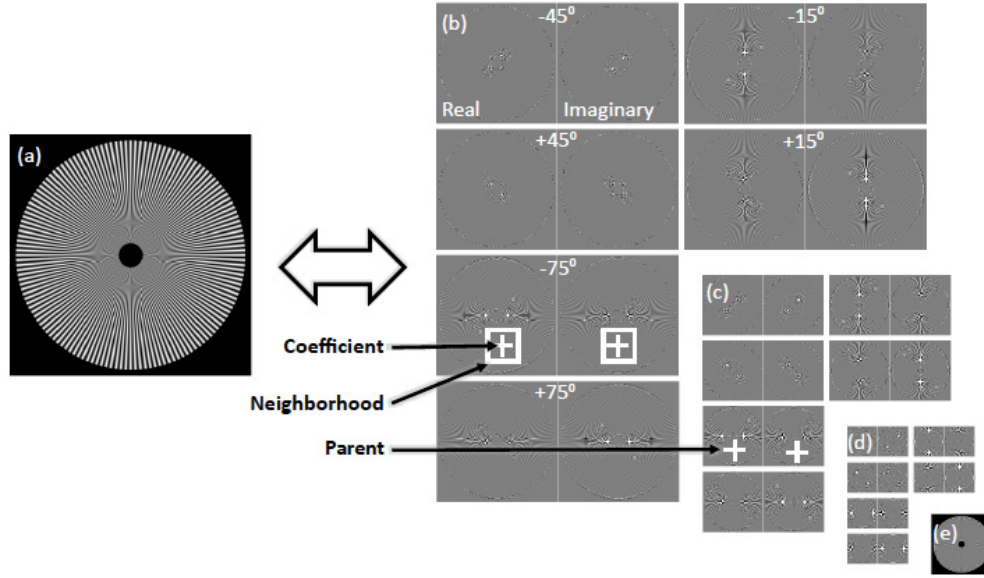
*junge@usc.edu

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Supplementary materials



Supplemental Figure 1. Schematics for Dual Tree-Complex Wavelet Filter (DTCWF). DTCWT results in two one real part and one imaginary part after the transform, with directional sensitivity for -75° , -45° , -15° , $+15^\circ$, $+45^\circ$ and $+75^\circ$ for each resolution level. (a) Original image before filtering. (b-e) The resulted filtered image of resolution level 2-5. The directionality of (c,d) is the same as image (b).

For the Dual Tree-Complex Wavelet Filter (DTCWF), We used LeGall 5, 3 tap filters for the first level and Kingsbury quarter sample shift orthogonal (Q-Shift) 10, 10 tap filters for higher levels [1] The filter coefficients are listed in Tables S1 and S2. The short filter banks have been chosen to limit the influence from neighbor pixels.

The wavelet coefficients $\Phi(l, b, x, y)$ are complex numbers for a particular level l , band b (1 to 6) and spatial coordinates x and y in the band. We used a BiShrink filter which shrinks the complex wavelet coefficient by considering the amplitude of the coefficient $\Phi(l, b, x, y)$, the amplitude of the parent coefficient $\Phi(l + 1, b, x/2, y/2)$, the variance in the neighborhood $\sigma_n^2(l, b, x, y)$ and a global estimate of the noise variance σ_g^2 [2].

The global noise estimate is determined by the median of absolute differences operator MAD considering the complex amplitudes of all coefficients in the two bands of the first level that have been horizontally and vertically high pass filtered (the two bands with directional sensitivity for -45° and $+45^\circ$):

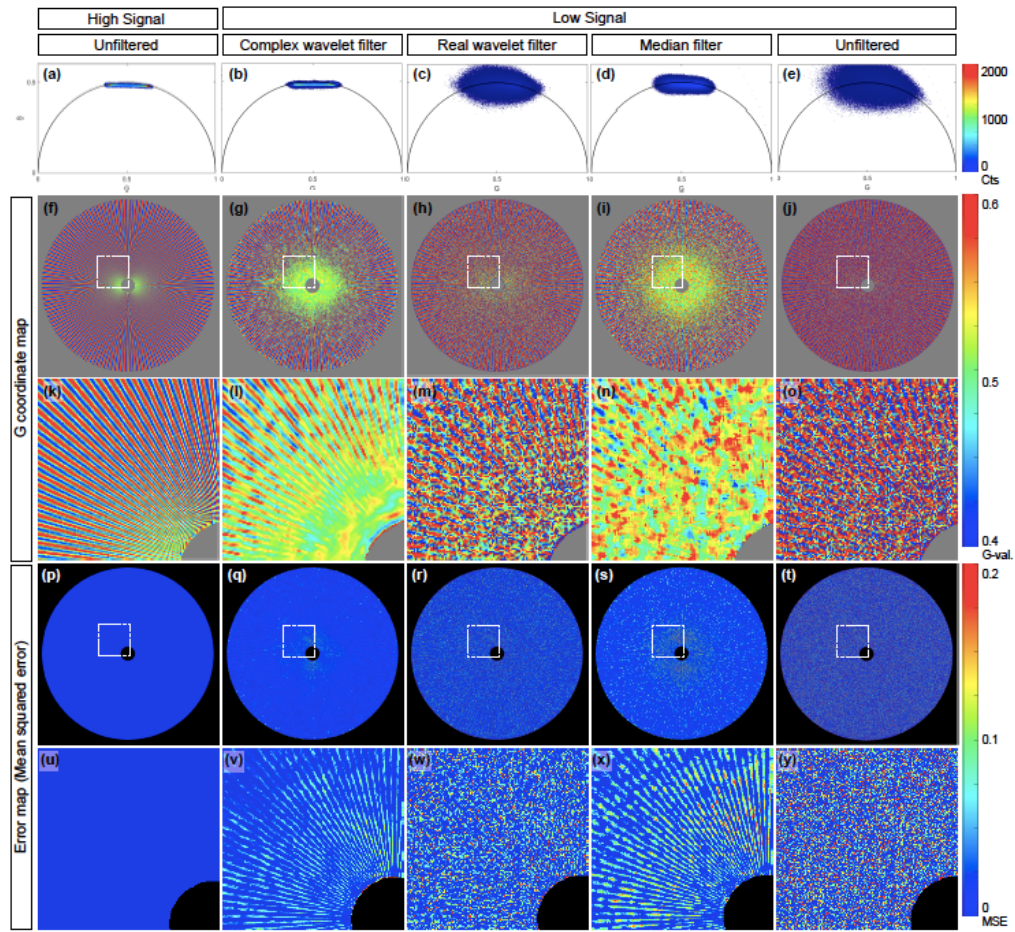
$$\sigma_g^2 = \frac{\text{median}(|\Phi(1, b, x, y)|)}{0.6745} \cdot \#(1)$$

A local noise estimate is calculated from the coefficients in the neighborhood N :

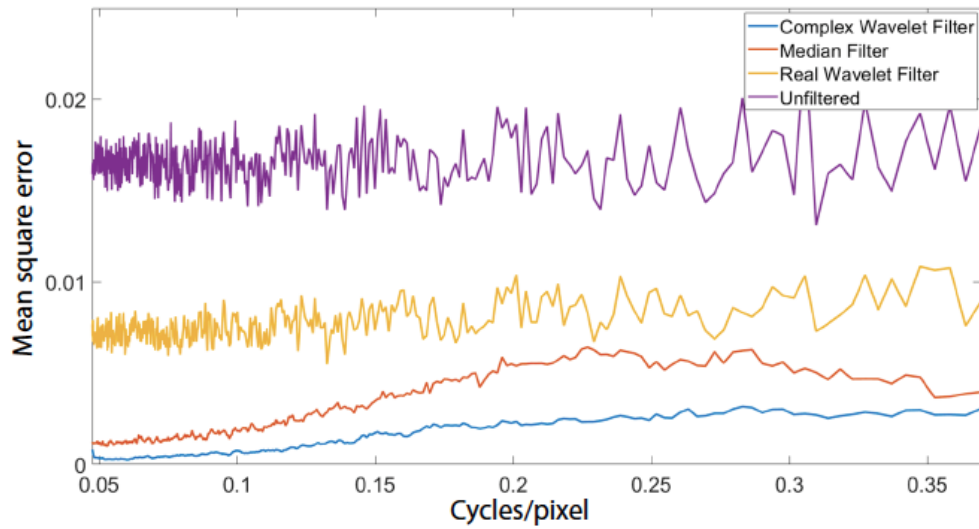
$$\sigma_n^2(l, b, x, y) = \frac{1}{(2N + 1)^2} \sum_{j=x-N}^{x+N} \sum_{k=y-N}^{y+N} \Phi^2(l, b, j, k) \cdot \#(2)$$

Note that all coefficients Φ in a band have a mean value of zero which simplifies the calculation in (1) and (2). We then calculated the filtered coefficients $\Phi'(l, b, x, y)$ based on the dependencies determined by Sendur and Selesnick [2, 3]:

$$\Phi'(l,b,x,y)=\left(1-\frac{\sqrt{3}\,\sigma_g^2}{\sqrt{\left\{\Phi^2(l,b,x,y)+\Phi^2\left(l+1,b,\frac{x}{2},\frac{y}{2}\right)\right\}\left(\sigma_n^2(l,b,x,y)-\sigma_g^2\right)_+}}\right)_+\Phi(l,b,x,y).\#(3)$$



Supplemental Fig 2. G-coordinate- and error maps of filtered simulated data. CWF achieves better structural reconstruction of FLIM phasor data than other filtering methods. FLIM phasor distributions of each image (a-e) show how similar the filtering methods (b-d) reassign FLIM signatures from the unfiltered, low signal image (e) compared to the ground truth (a). FLIM G-coordinate maps (f-j) are arranged in left-to-right order of increasing error (p-y) highlighting the detail of those errors in the zoomed in views (k-o) of f through j, respectively. Pixels with photon counts below 2 were thresholded out. Error maps (p-t) are generated by plotting the absolute difference of the filtered G coordinate maps with respect to the accumulated 50-frame ground truth. The color-scale bars show the colorimetric values of the G coordinate and Mean Square Error (MSE). Single-frame CWF was able to recover the most information, without introducing artifacts (b and l) and has the least error (q and v) of all the applied filtering methods. The real wavelet filter seems to have diminished detail in the G-coordinate map (c and m) but surprisingly has a similar error calculation (r and w) to the median filter (d, & n; o & x). The unfiltered single-frame G-coordinate map is extremely noisy, with heavy deviation from the actual G values in some regions (e) and, not surprisingly, has the most error (t and y). These data highlight CWF over other filtering methods across spatial frequencies, but especially the finer structures.



Supplemental Fig 3. CWF outperforms Median and Real Wavelet filters at all spatial frequencies. Mean Square Error (MSE) from simulated data in Supplemental Fig. 1 was calculated from the difference between the 1 frame accumulated data and the 500 frame accumulated data as ground truth. Unit for spatial frequency in the x-axis is cycles per pixel. All errors increase as spatial frequency increases but CWF MSE is lower than the MSE of both median and real wavelet filters. The decrease in MSE of the median filter above 0.3 cycles/pixel is due to artifacts in the ground truth simulation at high spatial frequencies.

Table S1. First level coefficients of the analysis filters

Tree A low-pass	Tree A high-pass	Tree B low-pass	Tree B high-pass
-0.125	0	0	0
0.25	-0.25	-0.125	0
0.75	0.5	0.25	-0.25
0.25	-0.25	0.75	0.5
-0.125	0	0.25	-0.25
0	0	-0.125	0

Table S2. Higher level coefficients of the analysis filters

Tree A low-pass	Tree A high-pass	Tree B low-pass	Tree B high-pass
0.0511304	-0.00618188	-0.00618188	0.0511304
-0.0139754	0.00168968	-0.00168968	0.0139754
-0.109836	-0.100231	-0.100231	-0.109836
0.26384	-0.000873623	0.000873623	-0.26384
0.766628	0.563656	0.563656	0.766628
0.563656	-0.766628	0.766628	-0.563656
0.000873623	0.26384	0.26384	0.000873623
-0.100231	0.109836	-0.109836	0.100231
-0.00168968	-0.0139754	-0.0139754	-0.00168968
-0.00618188	-0.0511304	0.0511304	0.00618188

Supplemental references

- [1] N. Kingsbury. "Complex wavelets for shift invariant analysis and filtering of signals," Appl. Comput. Harmon. A., 10, 234-253 (2001).
- [2] L. Sendur & I. Selesnick. "Bivariate shrinkage with local variance estimation," IEEE Signal Process. Lett., 9, 438 – 441 (2003).
- [3] L. Sendur & I. Selesnick. "Bivariate shrinkage functions for wavelet-based denoising exploiting interscale dependency," IEEE Trans. Signal Process. 50, 2744-2756 (2002).